Chronic Obstructive Pulmonary Disease (COPD)

SOURCE-WHO

Key facts

- Chronic obstructive pulmonary disease (COPD) is the third leading cause of death worldwide, causing 3.4 million deaths in 2023, approximately 6% of all global deaths *(1)*.
- Nearly 90% of COPD deaths in those under 70 years of age occur in low- and middle-income countries (LMIC).
- COPD is the seventh leading cause of poor health worldwide (measured by disability-adjusted life years).
- Tobacco smoking accounts for over 70% of COPD cases in high-income countries. In LMIC tobacco smoking accounts for 30–40% of COPD cases, and household air pollution is a major risk factor.

Overview

Chronic obstructive pulmonary disease (COPD) is a common lung disease causing restricted airflow and breathing problems. It is sometimes called emphysema or chronic bronchitis.

In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.

Smoking and air pollution are the most common causes of COPD. People with COPD are at higher risk of other health problems.

COPD is not curable but symptoms can improve if one avoids smoking and exposure to air pollution and gets vaccines to prevent infections. It can also be treated with medicines, oxygen and pulmonary rehabilitation.

Symptoms

The most common symptoms of COPD are difficulty breathing, chronic cough (sometimes with phlegm) and feeling tired.

COPD symptoms can get worse quickly. These are called flare-ups. These usually last for a few days and often require additional medicine.

People with COPD also have a higher risk for other health problems. These include:

- lung infections, like the flu or pneumonia
- lung cancer

- heart problems
- weak muscles and brittle bones
- depression and anxiety.

Common symptoms of COPD develop from mid-life onwards. As COPD progresses, people find it more difficult to carry out their normal daily activities, often due to breathlessness. There may be a considerable financial burden due to limitation of workplace and home productivity, and costs of medical treatment.

COPD is sometimes called emphysema or chronic bronchitis. Emphysema usually refers to destruction of the tiny air sacs at the end of the airways in the lungs. Chronic bronchitis refers to a chronic cough with the production of phlegm resulting from inflammation in the airways. COPD and asthma share common symptoms (cough, wheeze and difficulty breathing) and people may have both conditions.

Causes

Several processes can cause the airways to become narrow and lead to COPD. There may be destruction of parts of the lung, mucus blocking the airways, and inflammation and swelling of the airway lining.

COPD develops gradually over time, often resulting from a combination of risk factors:

- tobacco exposure from active smoking or passive exposure to second-hand smoke;
- occupational exposure to dusts, fumes or chemicals;
- indoor air pollution: biomass fuel (wood, animal dung, crop residue) or coal is frequently used for cooking and heating in low- and middle-income countries with high levels of smoke exposure;
- early life events such as poor growth in utero, prematurity, and frequent or severe respiratory infections in childhood that prevent maximum lung growth;
- asthma in childhood; and
- a rare genetic condition called alpha-1 antitrypsin deficiency, which can cause COPD at a young age.

COPD should be suspected if a person has typical symptoms, and the diagnosis confirmed by a breathing test called spirometry, which measures how the lungs are working. In low- and middle-income countries, spirometry is often not available and so the diagnosis may be missed.

Treatment

COPD isn’t curable, but it can get better by not smoking, avoiding air pollution and getting vaccines. It can be treated with medicines, oxygen and pulmonary rehabilitation.

There are several treatments available for COPD.

Inhaled medicines that open and reduce swelling in the airways are the main treatments.

Bronchodilator inhalers are the most important medicines for treating COPD. They relax the airways to keep them open.

Short-acting bronchodilators start to work in seconds and can last for 4–6 hours. These are often used during flare-ups.

Long-acting bronchodilators take longer to start working but last longer. These are taken daily and can be combined with inhaled steroids.

Other treatments may also be used:

- Steroid pills and antibiotics are often used to treat flare-ups.
- Oxygen is used for people who have had COPD for a long time or have severe COPD.
- Pulmonary rehabilitation teaches exercises to improve your breathing and ability to exercise.
- Surgery may improve symptoms for some people with severe COPD.

Some inhalers open the airways and may be given regularly to prevent or reduce symptoms, and to relieve symptoms during acute flare-ups. Inhaled corticosteroids are sometimes given in combination with these to reduce inflammation in the lungs.

Inhalers must be taken using the correct technique, and in some cases with a spacer device to help deliver the medication into the airways more effectively. Access to inhalers is limited in many low- and middle-income countries; in 2023 salbutamol inhalers were generally available in public primary health care facilities in less than two-thirds of low- and low-middle income countries.

Flare-ups are often caused by a respiratory infection, and people may be given an antibiotic or steroid tablets in addition to inhaled or nebulised treatment as needed.

Living with COPD

Lifestyle changes can help improve symptoms of COPD.

Quit smoking or vaping. This is the most important thing to do. Even if you have been smoking for many years, quitting can still help.

Avoid second-hand smoke or smoke from indoor cooking fires.

Stay physically active.

Protect yourself from lung infections:

- Get a flu vaccine every year.
- Get the pneumonia vaccine.
- Get all available COVID-19 vaccines and make sure you have had the latest boosters.

People living with COPD must be given information about their condition, treatment and self-care to help them to stay as active and healthy as possible.

WHO response

COPD is included in the WHO Global Action Plan for the Prevention and Control of Noncommunicable Diseases (NCDs) and the United Nations 2030 Agenda for Sustainable Development.

WHO is taking action to extend diagnosis of and treatment for COPD in a number of ways.

The WHO Package of Essential Noncommunicable Disease Interventions (PEN) was developed to help improve NCD management in primary health care in low-resource settings. PEN includes protocols for the assessment, diagnosis and management of chronic respiratory diseases (asthma and chronic obstructive pulmonary disease), and modules on healthy lifestyle counselling, including tobacco cessation and self-care.

WHO is currently developing new guidelines for the diagnosis and management of COPD in primary care.

Rehabilitation 2030 is a strategic approach to prioritize and strengthen rehabilitation services in health systems. Pulmonary rehabilitation for COPD is included in the Package of Interventions for Rehabilitation, developed as part of this WHO initiative.

Reducing tobacco smoke exposure is important for both primary prevention of COPD and disease management. The Framework Convention on Tobacco Control is enabling progress in this area as are WHO initiatives such as MPOWER and mTobacco Cessation.

Further prevention activities include the WHO Clean Household Energy Solutions Toolkit (CHEST) to promote clean and safe interventions in the home and facilitate the design of policies that promote the adoption of clean household energy at local, programmatic and national levels.

The Global Alliance against Chronic Respiratory Diseases (GARD) contributes to WHO's work to prevent and control chronic respiratory diseases. GARD is a voluntary alliance of national and international organizations and agencies from many countries committed to the vision of a world where all people breathe freely.

SOURCE-NIH

COPD, or chronic obstructive pulmonary disease, is a condition caused by damage to the airways or other parts of the lung. This damage leads to inflammation and other problems that block airflow and make it hard to breathe.

COPD can cause coughing that produces large amounts of a slimy substance called mucus . It can also cause breathing problems, shortness of breath, chest tightness, and other symptoms . Because COPD is a progressive disease, its symptoms often develop slowly but worsen over time and can limit your ability to do routine activities. Serious COPD may prevent you from doing even basic activities such as walking, cooking, or taking care of yourself.

The term COPD refers to two main conditions:

- **Emphysema** develops when there is damage to the walls between many of the air sacs in the lungs. Normally, these sacs are elastic or stretchy. When you breathe in, each air sac fills up with air, like a small balloon. When you breathe out, the air sacs deflate, and the air goes out. In emphysema, it is harder for your lungs to move air out of your body.
- **Chronic (long-term) bronchitis** is caused by repeated or constant irritation and inflammation in the lining of the airways. A lot of thick mucus forms in the airways, making it hard to breathe.

Most people who have COPD have a mixture of both emphysema and chronic bronchitis in different proportions, and how serious each condition is varies from person to person.

In the United States, COPD affects more than 14 million adults, and many others do not know they have it. More than half of those diagnosed are women. Rates are higher than average in American Indian and Alaska Native communities and in rural areas. COPD is a major cause of disability, and it is the sixth leading cause of death in the United States, according to the Centers for Disease Control and Prevention.

COPD symptoms

Recognize the Signs of COPD
Earlier diagnosis and treatment can make a big difference in quality of life for people who have COPD.

Common symptoms of COPD include:

- **Shortness of breath,** especially with physical activity. You may feel like breathing takes more effort or that you are gasping for air.
- **An ongoing cough** or a cough that produces a lot of mucus , sometimes called a smoker's cough. This is often the first symptom of COPD.
- **Wheezing or a whistling** or squeaky sound when you breathe.
- **Chest tightness or heaviness**, which may make it hard to take a deep breath or make it painful to breathe.
- Fatigue or extreme tiredness.

Not everyone who has these symptoms has COPD. Likewise, not everyone with COPD has all these symptoms. Your healthcare provider can determine whether you have COPD. You may also have

additional conditions that need to be diagnosed and treated. It is very common for people with COPD to have other conditions, including heart disease.

If your symptoms are mild, you may adjust your lifestyle to make breathing easier. For example, you may take the elevator instead of the stairs. For more serious symptoms, such as becoming short of breath during physical work, your healthcare provider can provide treatments and other ways to help. Severe COPD can cause additional symptoms, such as lack of appetite, loss of weight and muscle, anxiety, and depression.

When to call your healthcare provider

As soon as you notice symptoms of COPD in yourself or a loved one, make an appointment to see a provider. Don't wait for your symptoms to become serious.

A flare-up may require treatment in a hospital. You — or, if you are unable, your family members or friends — should **call 9-1-1** if you are experiencing unusual events:

- You are having a hard time catching your breath or talking.
- Your lips or fingernails have turned blue or gray, a sign of a low oxygen level in your blood.
- People around you notice that you are not mentally alert.
- Your heartbeat is very fast.
- The recommended treatment for your symptoms is not working.

How will your healthcare provider find out whether you have COPD?

Your provider will diagnose COPD based on your symptoms, your medical and family histories, and test results.

How Does A COPD Diagnosis Affect Your Loved Ones?
Learn what a COPD diagnosis means for your family, why it's important for caregivers to take care of themselves, and how other family members can be supportive.

- **Discuss your smoking habits** or whether you have had contact with secondhand smoke, air pollution, chemical fumes, or dust.
- **If you have persistent shortness of breath that gets worse over time,** let your provider know, because this is the main symptom of COPD.
- **If you have an ongoing cough,** tell them how long you've had it, how much you cough, and how much mucus comes up when you cough.
- Be sure to bring up any **family history of COPD**.
- **Your provider will examine you** and use a stethoscope to listen for wheezing or other abnormal chest sounds.
- **Lung function and imaging tests** will tell whether you have COPD and how serious it is.
- **If your provider recommends getting an X-ray or computed tomography (CT) scan** to detect lung cancer or diagnose a breathing problem, ask them about your risk of COPD.

- **If you get a COPD diagnosis, ask about genetic screening.** Some people have a genetic condition that increases their risk of COPD, called alpha-1 antitrypsin (AAT) deficiency. Screening is especially important if you have family members with AAT deficiency or who were diagnosed with COPD in their 40s or 50s or have liver disease. Likewise, if you are the first in your family to learn that AAT deficiency is part of your genetic makeup, ask your provider for advice on talking with family members about your results.

Spirometry

Diagnosing COPD
Learn about spirometry and other diagnostic tests your provider may use to diagnose COPD based on your symptoms and risk factors.

The main test for COPD is spirometry. It can detect COPD before symptoms are recognized. Your provider may also use the test results to find out how serious your COPD is and help set your treatment goals.

Spirometry is a lung function test that measures how much air you breathe out and how fast you can blow air out.

During the test, a technician will ask you to take a deep breath in. Then you will blow as hard as you can into a tube connected to a small machine called a spirometer. Your healthcare team may have you inhale (breathe in) medicine that helps open your airways and then blow into the tube again. They can then compare your test results before and after taking the medicine.